

Network Egress Blocklist and Exfiltration Attempt Analysis

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Introduction

This document provides a comprehensive reference guide for analyzing network egress activities related to suspicious and malicious exfiltration attempts. It catalogs known threat indicators including malicious domains, IP address ranges, ASN identifiers, detection signatures, and artifacts associated with recent incident response playbooks from the past six months. This guide aims to support threat-hunters, incident responders, and security analysts in identifying, analyzing, and mitigating potential data exfiltration by leveraging network logs, DNS data, and endpoint artifacts.

The scope covers detection signatures such as BEACON-12xx and C2-DNS-30xx, network indicators like c2-host-01.example.com and ASN-64512, as well as associated behaviors and artifacts such as mshta.exe executions, encoded PowerShell commands, and SHA256 hashes. The document also

includes procedures for analyzing logs, detecting anomalies, and executing containment and response operations.

Collection Overview

Our collection encapsulates threat-hunting and incident response playbooks compiled from the last six months of security operations. Key focus areas include:

- Detection signatures (e.g., BEACON-12xx, C2-DNS-30xx)
- EDR artifacts like process executions (mshta.exe, PowerShell)
- SHA256 hash repositories for known malicious payloads
- IAM audit logs indicating service account creations and scope anomalies
- Network blocklist entries: malicious domains, IP ranges, ASN identifiers
- Checkout failure codes: PAY-408, GATEWAY-52, CART-ERR-1203

The collection forms the basis for systematic threat detection, analysis workflows, and containment strategies.

Known Malicious Domains, IP Ranges, and ASN Identifiers

Malicious Domains

- **c2-host-01.example:** Command and Control host associated with suspicious traffic communication. Often resolves via DNS queries involving encoded payloads.
- **exfil-tracker.net:** Domain identified in exfiltration data flows, hosting malicious scripts.
- **suspicious-domain.org:** Related to beaconing activities detected during incident response.

Malicious IP Ranges

Start IP	End IP	Description
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192.168.100.0	192.168.100.255	Internal reconnaissance IP range used for lateral movement
203.0.113.0	203.0.113.255	Known exfiltration server hosting malicious payloads

ASN Identifiers

- **ASN-64512:** Belongs to malicious hosting provider known for facilitating exfiltration services.
- **ASN-65432:** Associated with VPN services often exploited for malicious communications.

Detection Signatures and Indicators

Threat Signatures

- **BEACON-1210:** Generic beaconing pattern detected via endpoint telemetry, often indicating backdoor activity.
- **C2-DNS-3035:** DNS queries matching known command and control patterns, such as encoded string payloads and unusual query types.

Endpoint Artifacts

Typical behaviors include:

- Execution of `mshta.exe` with encoded commands leading to remote command execution.
- PowerShell invocations with base64-encoded command parameters, e.g.:

```
PowerShell -EncodedCommand [Base64 string]
```

SHA256 hashes of typically observed payloads include:

```
a1b2c3d4e5f67890123456789abcdef1234567890abcdef1234567890abcde
...
```

Log Analysis Procedures

Prerequisites

- Access to NetFlow logs and DNS logs from relevant network segments
- SIEM or log analysis platform capable of filtering and pattern matching
- Endpoint telemetry data for process creation and command execution
- Threat intelligence feeds for malicious indicators

Step-by-Step Log Analysis

1. Identify abnormal DNS queries characteristic of C2 activity, such as high entropy strings or known malicious domains
2. Filter NetFlow data for outbound connections to suspicious IP ranges or ASN identifiers
3. Correlate DNS query timestamps with NetFlow connection logs to detect simultaneous activity
4. Verify process execution artifacts from endpoint logs, especially mshta.exe or PowerShell commands with encoded inputs
5. Compare SHA256 hashes of payloads with threat intelligence repositories to confirm malicious activity

Identifying Exfiltration Attempts

Indicators

- Unusual DNS queries to known malicious domains or high entropy string patterns
- Connections to IP ranges associated with known exfiltration servers or ASN identifiers
- High volume of outbound traffic from internal hosts to external hosts over standard ports

- Execution of suspicious processes such as `mshta.exe` or `PowerShell` with encoded commands
- Persistence of beaconing activity matching detection signatures (BEACON-12xx, C2-DNS-30xx)

Scenario Example

An internal host IP `10.0.0.45` generates DNS queries for `abc123.suspicious-domain.org` at regular intervals, matching C2-DNS-3035 signature. Simultaneously, on endpoint logs, process `powershell.exe` is observed executing with a base64 encoded command. NetFlow logs reveal outbound connections to IP `203.0.113.45`, within the malicious IP range. SHA256 hashes of payloads match known malicious hashes in threat intel feeds.

Containment and Response Procedures

Initial Containment

- Immediately isolate affected hosts from the network
- Block outbound traffic to known malicious domains and IP ranges at the firewall level
- Disable suspicious service accounts involved in exfiltration activities (e.g., accounts with auto-creation events or with SCOPE-PROD-API-ADMIN scope)

Investigation Steps

- Collect endpoint artifacts, including process execution logs and payload hashes
- Analyze DNS query patterns and NetFlow data for additional compromised hosts
- Confirm exfiltration channels and data volume

Remediation

- Remove malicious files and scripts from affected hosts
- Implement targeted firewall rules to prevent further data exfiltration
- Revoke and rotate compromised credentials and API keys

API Throttling Playbook

Objective

Limit the rate of API requests during suspected exfiltration to prevent data theft or abuse, and to enhance detection accuracy.

Procedure

1. Identify critical APIs related to data access (e.g., data export endpoints)
2. Configure throttling policies in the API gateway (e.g., limit requests to 10 per minute per user/IP)
3. Monitor for anomalies or spikes in API usage
4. Notify security team of potential abuse or breach

Monitoring

Use threat intelligence to adjust thresholds dynamically and track suspicious API patterns during investigations.

Partner OAuth Review Procedures

Purpose

Conduct thorough reviews of third-party OAuth integrations to prevent unauthorized data sharing or exfiltration.

Review Steps

1. Audit all active OAuth clients and their scopes, focusing on those with API scope IDs like `SCOPE-PROD-API-ADMIN`
2. Verify the purpose and necessity of each client, especially third-party applications
3. Assess recent activity logs for abnormal access patterns or elevated privileges
4. Revoke access tokens associated with suspicious or dormant clients
5. Implement minimum privilege principles and scope separation

Best Practices

- Regularly review OAuth scopes and permissions
- Implement Multi-Factor Authentication (MFA) for critical OAuth clients
- Use audit logs to trace OAuth token issuance and usage

Appendices

Sample Detection Signature Log Entry

```
Date: 2024-03-15 14:23:56
Signature: C2-DNS-3035
Host: internal-host-07
DNS Query: abc123.suspicious-domain.org
Outgoing IP: 203.0.113.45
Payload SHA256: a1b2c3d4e5f67890123456789abcdef1234567890abcde
Process: powershell.exe
Command Line: PowerShell -EncodedCommand V2luZG93cy4u...
Status: Suspicious activity confirmed
```

Sample Log Analysis Checklist

- Verify DNS query entropy and frequency
- Check for outbound connections to malicious ASN IP ranges
- Assess process trees for command lineage
- Cross-reference payload hashes with threat intelligence